

**Master of Computer Application (MCA) (Two Yrs Programme) - 3rd
Sem. (Batch 2024-26) (CBECS)**

(2225)

Paper: CSL5074 Cloud Native Application Development

Time allowed: 3 hrs.

Max. Marks: 70

Note: Students are required to attempt five questions, selecting at least one question from each section. The fifth question may be attempted from any section. All questions carry equal marks.

Section - A

- Q1.** (a) What are the main characteristics of cloud computing and explain how it meets needs like scalability, cost-efficiency and flexibility. (7)
(b) What are the pros and cons of adopting cloud computing? (7)
- Q2.** Explain the different types of Cloud Deployment Models and Service Models, how organizations apply these models in real-world scenarios to meet various business and technical needs. (14)

Section - B

- Q3.** (a) Define Virtualization and explain why it is essential for modern IT infrastructures. Describe any three types of virtualization with suitable examples. (7)
(b) Explain the mobility patterns V2V, V2P, P2P, D2C and D2D in virtualization. What does each pattern represent? (7)
- Q4.** What is a Hypervisor? Describe its role in enabling virtualization. Compare the two main types of hypervisors in terms of architecture, performance, and real-world applications. (14)

Section - C

- Q5.** What is Docker? Explain the architecture of Docker in detail, describing the role of its main components. Also, differentiate between a container and a virtual machine. (14)
- Q6.** (a) Describe any ten essential Docker commands including their syntax, purpose and common use scenarios. (7)
(b) Explain how Docker Swarm architecture enables high availability. How does Docker Swarm manage and maintain services to ensure reliability? (7)

Section - D

- Q7.** Explain Kubernetes and discuss how its architecture and features enable effective orchestration of containerized applications. (14)
- Q8.** What is a configuration management tool and discuss its importance in automating IT operations. Explain how Chef's Master-Slave architecture supports configuration management. (14)
